Where authority resides: A completion-based taxonomy of tokenisation

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Abstract

Systems described as "tokenisation" perform fundamentally different institutional functions yet are analysed as though they denote a single economic transformation. This categorical confusion produces contradictory empirical findings across liquidity, settlement, risk, and governance in tokenised markets. This paper proposes a completion-based taxonomy that classifies tokenised arrangements according to when entitlement becomes operative and which institution determines that operability. Four distinct categories emerge. Referential tokenisation records exposure while completion depends on external redemption. Contractual tokenisation activates enforceable rights whose effect depends on legal or administrative action. Title tokenisation transfers recognised ownership within an institutional framework accepting the recorded state as authoritative. Atomic tokenisation completes transfer through system operation alone. The framework generates four testable predictions, each observable in current markets. Liquidity is inversely proportional to real-world finality embedded in the transfer. Systems where transfers do not complete entitlement cannot function as settled base assets. The visible reliability of ledgers masks invisible variability of authority, producing systematic risk misclassification. Governance without jurisdictional reach is coordination, not control. Token standards evolved by progressively internalising institutional functions, moving unintentionally from accounting to identification to validation to recognition. The atomic tokenisation of real-world assets remains elusive because authority structures remain external to records. The contribution is a single organising variable, the location of completion authority, that reconciles previously contradictory observations across these disciplines.

Keywords: *tokenisation, blockchain, completion authority, settlement finality, property rights, institutional economics, digital assets, distributed ledger technology*

Tokenisation dominates blockchain discussions (BIS, 2023; OECD, 2024). Sovereign actors see tokenising infrastructure and national assets as new funding methods (Garrido, 2023). Financial institutions consider tokenised securities the next step in capital markets. Technology platforms view tokenisation as the settlement layer for digital commerce and automated agents (CFA Institute, 2025; FSB, 2024). These applications are broad and cause problems. Classifying different meanings, functions, and structures as a single category is misleading. Shared language implies that tokenisation means a single economic change. It does not.

Systems, structures, and methodologies grouped under "tokenisation" do not perform a common function but instead reflect distinct institutional roles performed by the token itself (FSB, 2024; Garrido, 2023). Existing taxonomies typically classify tokenisation by issuer structure, legal wrapper, or asset type, whereas this paper adopts completion of entitlement as the organising variable, distinguishing arrangements by when transfer becomes operative rather than by how they are structured. The distinction between these roles depends on four structural questions. What moves when a token moves? When does entitlement become operative? Which institution determines the operability? Is the transfer completed by record or by recognition? A token that provides price exposure to an asset is fundamentally different from a token that represents participation in a contract (FSB, 2024). Both also differ from a token used as a settlement instrument or recorded in a registry (BIS, 2023; Zetzsche et al., 2020). Concrete examples illustrate this distinction. There is a fundamental difference between a real estate project that assumes each property is held by a separate Decentralised Autonomous Organisation (DAO) that tokenises participation whilst the underlying legal title to the property is never visible to token holders, and a YouTube channel that tokenises its future advertising revenue stream, or offshore special purpose vehicles (SPV) that tokenise the value appreciation of luxury goods or potential winnings of elite athletes or racehorses. The design of each token varies by asset class, use case, economic construction, and how they create or transfers rights, or change entitlements between parties, within a given legal framework (Zetzsche, et. al., 2020; Schmidt & Wagner, 2019). Some tokens relate to ownership, some limit it, and others claim to constitute it.

Treating these mechanisms as a single category creates persistent analytical contradictions, as each discipline draws internally consistent but mutually incomparable conclusions from structurally different arrangements (FSB, 2024; OECD, 2024). This confusion creates risks that can have serious immediate and long-term market effects (BIS, 2023; Teixeira, 2025). Financial markets welcome the settlement efficiency and liquidity expected from tokenisation (CFA Institute, 2025), while market analysts identify liquidity shortfalls (FSB, 2024; Mafrur, 2025; OECD, 2024). Legal scholarship of tokenisation has focused on enforceability, custody limits, and jurisdictional differences (Werbach & Cornell, 2017; Reyes, 2020) while financial and technical literature describes tokenisation as transforming ownership infrastructure (Zetzsche et al., 2020; Biais et al., 2019). Each field's conclusions are consistent internally but remain self-referential and tautological.

"Tokenisation" groups systems that record economic relationships, restrict transfers, and claim to define them (Hodgson, 2015; Zetzsche et al., 2020). Failing to distinguish categorical differences leaves analysis descriptive at best. Without a shared ontology, meaningful cross-disciplinary comparison remains blocked, risks stay obscured, regulatory alignment falters, and broad negative financial impacts become more likely. More importantly, this obscures risk, hinders regulatory alignment and enforcement, and can cause broad negative financial impact.

Economic ownership is not simply passive investment because it can include rights and responsibilities. Owners face audit, tax, regulatory, and insolvency consequences. For productive assets, ownership brings employment obligations, insurance, and legal compliance. Economic ownership is subject to both gains and losses. Asset ownership includes the capacity and right to manage it, more than value accumulation (Besley & Ghatak, 2010).

Legal ownership requires recognition by formal legal systems that go far beyond a basic understanding of possession. The title exists only when the economic relationship appears in legal records, such as a company or land register. Documented holders in the system can assert claims, exercise rights, and carry liabilities. Exposure to an asset's value or fractional benefits is not ownership. Title, governance, and responsibilities may still be held by other, possibly unknown, entities. System recognition confers ownership, control, and goes far beyond passive fractionalised involvement (Hodgson, 2015; De Soto, 2000).

A ledger is any system of record that permanently tracks changes in relationships between parties, the form of which does not matter. It can be a central database, a distributed technical system, or paper. A ledger matters because it records state changes so others can check, verify, and trust them. If the ledger records a transfer, it seals that action into a recognized event. This entry may trigger other actions, like moving funds or updating a register. The record does not itself complete the transaction, but simply documents it. Law determines if and how the recorded entry makes the transfer valid. In conventional settings, a contract serves as a record and provides legal protection. When parties sign a contract, the agreement is complete. A ledger event just adds documentation to the history. The entry confirms a legally valid transfer but does not complete it. Only in rare cases does the ledger itself complete the transfer (BIS CPMI, 2017; Liao, 2020). In insolvency or when an intermediary fails, the distinction between a record and its operative effect becomes materially consequential. Depending on where completion authority resides, the holder's position may shift from a proprietary interest to that of a general unsecured creditor. Completion of an asset's transfer does not imply completion of the payment leg, as authority over the settlement medium introduces an additional layer of operability that sits beyond the scope of this taxonomy.

Authority over a transaction comes from the legal framework, not the record. A ledger entry matters only if legal and social systems treat it as the point at which rights change. Like money, a record gains power when consensus says it settles facts. If economic, legal, and recorded states align, the transfer is self-contained. Trust comes from execution. Validity and settlement are simultaneous. No later checks or reconciliation are needed. Confidence comes from transactions where recognition, execution, and record all match (Wandhöfer & Berndsen, 2019).

Tokenisation's significance depends on the extent to which a recorded transfer relies on external institutions after the ledger event. In weaker forms of tokenisation, the ledger simply records an action that still requires legal, administrative, or custodial validation, enforcement, or reconciliation. In stronger forms, the ledger event alone is sufficient to fulfil settlement and recognition criteria. This variance is caused by the relocation of authority from external institutions into the system and not the technology of tokenisation itself. The fundamental change exists beyond the nature of the transaction in when and how legal and economic effects become operative (Liao, 2020; Wandhöfer & Berndsen, 2019).

The distinction discussed is practical and central to tokenisation analysis. The term "tokenisation" is applied to systems that serve different institutional functions (IOSCO, 2025). For example, some market participants may believe they own assets when they have only price exposure, regulators may apply ownership-based frameworks to systems that provide only references, technologists may claim settlement finality when only a record exists (CFA Institute, 2025; Mirdala, 2025). Such mismatches are categorical problems that exist beyond the technology. Assessing risk and enforceability requires determining who holds authority, which differs by implementation and jurisdiction (BIS CPMI, 2017; European Commission, 2025; Liao, 2020). Tokenisation should not be treated as a single category but classified according to when a transfer is legally operative.

If completion timing changes, the transaction changes. Describing different systems with the same term causes indifference and ignores distinctions. As a result, liquidity, ownership, settlement, and risk are wrongly treated as a single mechanism despite distinct origins. The central question must focus on when the transfer is complete and not simply consider what the token itself signifies (Liao, 2020; BIS CPMI, 2017).

This paper examines property theory, market microstructure, and institutional economics. These perspectives are integrated through a comparative framework that traces how different institutional processes determine when entitlements become operative. The analysis applies property theory to identify distinctions between title, beneficial interest, contract claims, and register roles in determining when transfers become binding (Honoré, 1961). From financial economics, it draws on market microstructure to distinguish trade execution, clearing, and settlement, and to define concepts such as irrevocability (European Settlement Finality Directive, 1998; Liao, 2020; BIS CPMI, 2017). Institutional economics examines how shifts in authority among registrars, issuers, courts, custodians, and protocols shape the category and effectiveness of tokenisation arrangements (North, 1990; Hart, 1995). By explicitly mapping these disciplinary lenses to specific elements of tokenised systems, the analysis triangulates between legal, financial, and institutional contexts to support a rigorous, interdisciplinary framework.

Most writing on tokenisation examines issues within disciplinary silos. Legal scholarship covers enforceability, custody, and jurisdictional legitimacy (Werbach & Cornell, 2017). Financial research covers liquidity, settlement efficiency, and market design (CFA Institute, 2025; Aldasoro et al., 2023). Technical literature discusses programmability and composability (BIS, 2024). Each perspective is internally consistent, but the conclusions seem contradictory because the analysis is inconsistent. This paper’s contribution is to specify the object of analysis. Tokenised arrangements vary depending on when the entitlement becomes operative and which institution determines it.

By making completion authority the organising variable, this analysis presents a framework for bringing together these bodies of work without reducing tokenisation to either utopian claims of on-chain finality or simplistic critiques of market underperformance. This analysis seeks to transcend the repetitive critiques often associated with perceived failures of tokenisation by proposing a taxonomic framework. It aims to avert the assessment of tokenisation through inappropriate equivalence, rather than to challenge its feasibility or long-term promise. By separating what is represented from what is transferred, the paper deals with a persistent source of common misunderstanding. Many failures attributed to tokenisation arise from the misidentification of the nature of the entitlement being transferred, rather than from the transfer mechanism itself (Mirdala, 2025).

The distinctions developed in this paper can be previewed in simplified form below to provide a conceptual grid as the following content is introduced. Tokenised arrangements differ not by asset class or technology, but by what moves when a token is transferred, and which institution makes that movement operative. The detailed analytical framework and implications are developed in the sections that follow.

Table 1: Conceptual overview of completion authority across forms of tokenisation

Tokenisation Form	What Moves	Ledger Role	Example	Completion authority
Referential	Exposure	Mirror	Stablecoin, synthetics	Honour/performance
Contractual	Enforceable Claim	Execution Trigger	STO, funds	Legal enforcement
Title	Legal Ownership	Official Register	DLT Securities	Legal recognition
Atomic	Ownership itself	Mechanism of Transfer	Native cryptographic assets	System operation

The Spectrum of Tokenisation

Tokenisation can be arranged on a continuum based on the point at which a transaction is deemed complete, signifying the instant at which no additional institutional action is necessary for the alteration in rights or value between the parties to be realised. In this taxonomy, the "ledger" is a digital record of transactions on a blockchain. If another actor needs to validate, enforce, settle, or recognise a transaction, then recording a ledger event, agreeing to terms, or signalling intent does not complete it. Completion, therefore, refers to the point at which the change becomes operative in practice. This exceeds the basic description of a promise. The determining factor is whether the ledger describes, activates, recognises, or constitutes a transfer.

A useful taxonomy must do more than organise language. It must predict outcomes. If the classification proposed here is accurate, arrangements located at various stages of completion ought to demonstrate consistently divergent behaviour. Systems where completion remains external to the transaction should demonstrate persistent reliance on counterparties, discount pricing, and constrained composability. Systems where transaction completion is internal should demonstrate final settlement, autonomous composability, and be jurisdictionally agnostic.

Empirical evidence supports these distinctions. For instance, fiat backed stablecoins such as Tether (USDT) and USD Coin (USDC), which operate in the referential tokenisation category, have historically traded below their peg during periods of redemption stress, reflecting counterparty risk and reliance on issuer performance (Diop, et. al., 2024). In the contractual category, tokenised private equity funds and real estate shares, such as those issued via platforms like tZERO or RealT, exhibit discounted secondary market pricing and low liquidity compared to direct ownership or listed securities, illustrating constraints imposed by required enforcement and legal process (FSB, 2024; OECD, 2024). Discounts to net asset value are not unique in tokenised structures and are also observed in non-tokenised asset pools where they reflect valuation timing, liquidity constraints, or market expectations. However, in tokenised contractual arrangements an additional dimension is possible. The market prices the externality of the entitlement with the related uncertainty when the effectiveness of the transfer is dependent on governing agreements rather than the ledger state alone. In the title category, legally recognised digital registers for securities, where the ledger is accepted as the authoritative ownership record within a governing legal framework, reduce enforceability discounting because recognised title updates when the authorised record updates, although liquidity often remains constrained by permissioning, eligibility rules, and jurisdictional recognition requirements (Zetzsche et al., 2020). By contrast, native digital assets such as Bitcoin and Ether, which are examples of atomic tokenisation, display autonomous composability and jurisdictional agnosticism. These transfers are final at the point of recording, allowing direct use as collateral, settlement asset, or programmable resource in DeFi protocols without reference to external authority or reconciliation (BIS, 2023). These divergent behaviours across categories substantiate the taxonomy's predictive value. The framework's validity is based on the consistent manifestation of these differences in practice and not subjective interpretations.

Referential tokenisation

Referential tokenisation, such as stablecoins or synthetic exposure instruments, is where the ledger records exposure, while the change in economic position between the parties is recorded elsewhere. The holder's gain or loss only becomes real when an external actor honours a redemption, settles a payment, or performs an obligation linked to the token (Garrido, 2023; BIS, 2020; Zetzsche et al., 2020). The holder does not enter the economic structure of the underlying asset. Instead, they enter a relationship with an issuer, counterparty, or market mechanism that references it. Profit and loss depend on whether that reference continues to perform or be honoured. It is not based on whether the holder participates in the asset's governance, cashflow rights, or liabilities. The referential token provides economic exposure without economic involvement.

The ledger records balances or exposure, while ownership, settlement, and enforceability remain external to the system. The token operates as a representation or mirror of value rather than an instrument of transfer. Completion occurs only when institutional actors outside the ledger recognise and settle the underlying relationship (BIS, 2020; Garrido, 2023). Typical characteristics include price exposure without legal title, redemption dependent on issuer action, and market trading independent of asset governance.

Contractual tokenisation

Contractual tokenisation which can best be observed in tokenised fund interests, or special purpose vehicle participation rights, is where the ledger records a change that activates enforceable rights with enforceable consequences for each party. The change in entitlement between parties is determined by an external legal contract or document. The holder's position becomes real through the capacity of governing documents to compel performance outside the ledger transaction. It does not depend on voluntary counterparty action. The holder enters into a defined legal relationship with the underlying asset, but ownership remains dependent on interpretation and enforcement outside the ledger. Profits, losses, and participation come from contractual obligations rather than the ledger state itself.

The ledger records the event signalling performance in fulfilling the obligation created under the agreement, while validity, settlement, and recognition remain functions of external administrative or legal processes. The token serves as an execution mechanism within the contractual structure rather than the definitive record of title. Completion occurs when the agreement is recognised and enforceable in law, not when it is entered in the ledger. Typical characteristics include legally enforceable claims without formal title, transfer subject to approval or compliance checks, and disputes resolved through contractual interpretation rather than ledger state.

Title tokenisation

Title tokenisation occurs in distributed ledger technology (DLT) recorded securities or in registered digital shares and happens when the recorded state of a transaction is seen as the official source of ownership by a governing body. The change in entitlement takes effect because the holder is entered into the title record itself, not because they can compel performance under a contract. The holder's position depends on recognition within the institutional system that determines valid title. There is no appeal to enforcing obligations against a counterparty because those obligations are embedded in the title itself.

The ledger forms part of the mechanism through which ownership is recognised. Transfer restrictions relating to identity, compliance, or jurisdiction are imposed as prerequisites for recognition rather than prerequisites for performance. The transaction is resolved at the point the record is acknowledged as the definitive statement of ownership, even while the authority of that record continues to stem from the external legal or administrative framework regulating the title system (Zetzsche et al., 2020). Profit, loss, and participation emerge from the holder's position within the asset structure. The holder is treated as the party to whom both the interest and performance of the asset belong. Typical characteristics include permissioned transfer, identitybound holding, and recognition of the record as the official statement of ownership within a legal system.

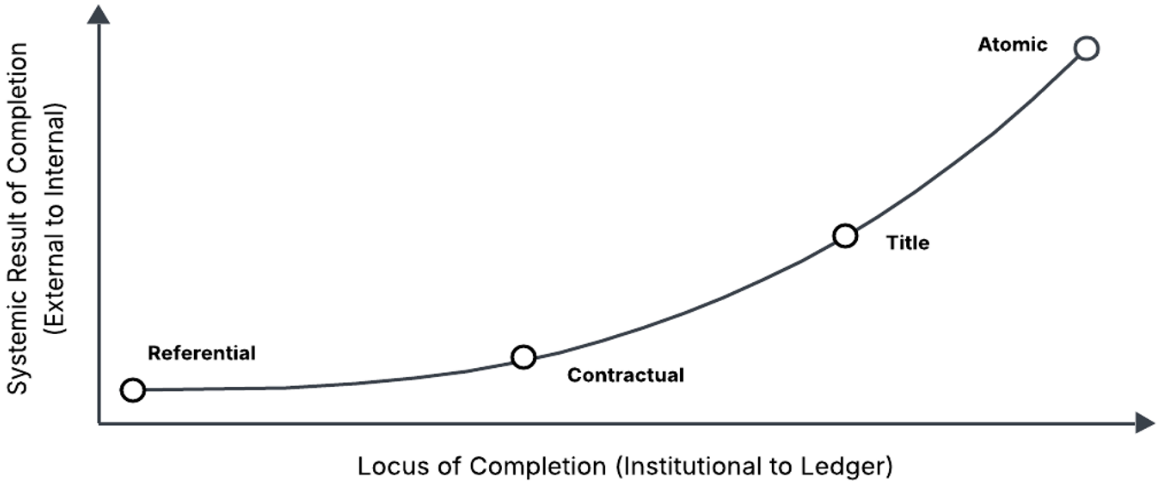
Atomic tokenisation

Atomic tokenisation is best depicted in native crypto currencies like Bitcoin and occurs when the recorded state itself completes the transfer of entitlement between the parties. No more validation, reconciliation, or institutional confirmation is required for the change to take effect. The ledger is more than the evidence of a transaction nor part of a recognition process. It is the mechanism through which ownership changes. Economic, legal, and recorded states coincide at execution, so the moment of recording is the moment the transfer becomes operative.

Here, the governing framework treats system operation as sufficient for validity. The holder does not rely on contractual enforcement or institutional recognition because the conditions required for transfer are embedded within the system operation itself. Authority determining ownership is internal to the recording mechanism rather than external to it. Disputes concern whether the system operated correctly rather than whether an external obligation was honoured or recognised. The transaction is resolved by reference to system state, not by interpretation, approval, or registration outside it. Typical characteristics include final transfer at recording, no external reconciliation, and authority embedded in system operation.

These four types of tokenisation are distinguished by institutionally discrete sources of authority. Each arrangement locates a different point at which transfer is considered complete. In referential systems, completion depends on honouring, in contractual systems, on enforcement, in title systems, on recognition, and in atomic systems, on operation itself (Garrido, 2023; BIS, 2020; Liao, 2020; Wandhöfer & Berndsen, 2019). The same specialized instrument can occupy different positions along this continuum depending on how surrounding rules treat the recorded state (Zetzsche et al., 2020). Hybrid or ambiguous structures frequently arise, especially as new technical or regulatory models emerge. For example, some implementations combine contractual and title features or shift position as institutional arrangements evolve. In such cases, the taxonomy is applied by examining the specific location where authority to make entitlement operative resides at each point in the process. When systems overlap categories or transition among them, classification should reflect the most authoritative institutional dependency that remains after the ledger event. This acknowledges that tokenisation is not always cleanly separated into discrete types. Tokenisation exists on a spectrum with blurred boundaries. The opportunity of the technology is thus a set of arrangements defined by where authority to determine entitlement resides, rather than a single, homogeneous taxonomic category. The following section applies this framework to existing implementations to classify them according to when the transfer becomes operative.

Figure 1: Internalisation effects across the locus of completion



Note. The figure illustrates the relationship between the locus at which completion occurs and the systemic result that follows. As the point of completion moves from external institutional enforcement toward ledger-native execution, the effects of completion become increasingly internal to the system. Referential arrangements rely on external validation and produce minimal internal settlement effects. Contractual structures introduce enforceable linkage but remain externally dependent. Title based systems shift recognition toward the ledger. Atomic arrangements complete entitlement at the moment of ledger state change, producing fully internalised settlement effects. The curved relationship reflects the non-linear nature of this transition whereby behavioural consequences accelerate as completion becomes native to the ledger.

Applying the framework

The framework above allows tokenised systems to be classified by when entitlement becomes operative, rather than by technological architecture, asset class, or regulatory label. The analysis prioritises what must happen following a ledger event for the holder's position to be validated, rather than sole confidence in the token's representation. Where further action, enforcement, recognition, or interpretation is required, the system occupies an earlier position in the taxonomy. Where no further institutional step is required, it takes its place later. The same asset and identical smart contract logic may therefore fall into different categories depending on the governing rules surrounding the record.

Industry descriptions of tokenisation frequently classify systems by asset type, regulatory treatment, or technical standard. These labels suggest similarity where the underlying mechanisms diverge and suggest difference where the operative structure is identical. The framework developed throughout instead classifies implementations based on when entitlement becomes effective. The result is that commonly grouped systems separate, while apparently unrelated systems converge.

The subsequent analysis of real-world examples examine what must occur after a transfer is posted to a holder's position for it to materialise. If performance, interpretation, or recognition still needs to occur, the system is at a lower level of the taxonomy. If the recorded state itself settles the transfer, it is in a later one.

Referential tokenisation: Stablecoins

Fiat backed stablecoins are commonly described as payment instruments because transfers between users settle immediately on the ledger. However, under this proposed framework, the relevant question is whether the change in entitlement is finished rather than the transfer being finalised. A ledger transfer moves the token, but the holder does not receive the referenced currency right away. The holder instead acquires a claim against the issuer that can be exercised in accordance with the issuer's redemption process (Garrido, 2023; J.P. Morgan, 2025).

To convert the token position into the referenced value, the holder must meet several prerequisites. These include submitting a redemption request, satisfying compliance checks, and relying on the issuer to perform settlement through conventional payment infrastructure (MIT DCI, 2025). The change in entitlement only happens when the external settlement is carried out, even if the issuer is operating correctly and there are no problems. If performance fails, the holder's remedy lies in legal enforcement rather than in the recorded state (Garrido, 2023).

The system, therefore, separates two events. The movement of the token and the completion of the underlying transfer. The ledger conclusively resolves the former but not the latter. The stablecoin serves as a circulating representation of a claim whose satisfaction depends on performance outside the system. According to the proposed taxonomy, completion relies on honouring rather than enforcement, acknowledgement, or operation. Fiat backed stablecoins thus occupy a referential position whereby the token settles internally while the referenced entitlement settles externally. Therefore, in practice, issuers retain the capacity to burn, freeze, or reassign tokens in response to compliance, custodial, or legal instructions. This demonstrates where entitlement remains contingent to off-ledger authority. For instance, Circle's USDC involves such compliance checks during redemptions (Garrido, 2023).

Contractual tokenisation: Securities and funds

Tokenised securities and fund interests are frequently presented as direct digital ownership of underlying assets. The ledger tracks the allocation of units and enables transfer between holders, giving the appearance that ownership moves with the recorded state. Under the current framework, the focus moves to whether the change in entitlement takes effect when it is recorded.

In typical implementations, the holder's rights arise from subscription agreements, operating agreements, or share instruments that exist independently of the ledger (SEC, 2026). The token transfer signals a change that must be recognised and effected by administrators, transfer agents, or legal custodians who keep the official ownership records (IOSCO, 2025). Where required conditions are not satisfied, the ledger state alone does not determine valid ownership. The holder therefore acquires enforceable rights rather than a recognised title. If a dispute arises, resolution depends on interpreting the governing documents and enforcing performance by the responsible parties, rather than treating the recorded state as conclusive (SEC, 2026; Osler, 2025). The ledger initiates commitments but does not resolve the transfer itself.

Because performance is not voluntary, these systems differ from referential arrangements. The holder may compel recognition of their position through legal enforcement, yet they likewise differ from title systems because the authoritative ownership record exists outside the ledger and may override it (IOSCO, 2025). Completion is contingent on enforcement rather than adherence or acknowledgement. Tokenised securities and fund interests, therefore, occupy the contractual position in the taxonomy where the ledger initiates rights, whose effects materialise through enforcement of the governing agreement. For example, tZERO platform's secondary market pricing often shows discounts due to these constraints (FSB, 2024).

Title tokenisation: Legally recognised digital registers

A different structure arises where the token transfer modifies the legally recognised ownership record itself. In these systems, the ledger does not instruct an administrator to change ownership, and the holder does not rely on enforcing contractual rights against a counterparty. The ownership record maintained by the issuer or authority is defined as the recognised record produced by the system, so when the recorded holder changes, the recognised owner changes.

This structure is already present in jurisdictions that recognise digital registers as official title records. Under the Swiss DLT securities framework (Registerwertrechte), a company may designate a blockchain register as its legal share register (CMTA, 2020; Lavayssière, 2025). A transfer recorded in the authorised ledger, therefore, constitutes a change in shareholder ownership rather than initiating a later assignment or requiring administrative approval. A dispute concerns whether the transfer complied with the register rules, not whether a party must perform an obligation.

An equivalent structure exists under the German Electronic Securities Act (eWpG), where securities may exist solely as entries in an approved electronic register, including crypto based registers (BaFin, 2026). The legal status of the security follows the register entry itself. Courts determine ownership by reference to the recognised record rather than by enforcing a separate contractual transfer. Pilot land and asset registry systems in several jurisdictions, including the United Arab Emirates, explore the same model. A transfer becomes effective because the authoritative record changes, not because an intermediary subsequently honours or executes it.

This title tokenisation is distinct from contractual tokenisation in these examples is that the holder already possesses recognised ownership and does not need to compel any third party to provide it. The distinction from atomic tokenisation is that authority still derives from the legal framework that designates the register as authoritative rather than from the recording mechanism alone. Completion, therefore, depends on recognition of the record within an accepted governing framework. These systems occupy the title position in the taxonomy where entitlement changes when the recognised ownership record changes.

Atomic tokenisation: Native digital assets

The final category describes an arrangement in which the recorded state itself completes the transfer of entitlement, and no external institution is required to validate, recognise, or enforce the change. Economic, legal, and recorded states coincide at execution. The transfer becomes operative because the system treats the change of record as the change of ownership.

This condition exists today only for assets native to their own recording system. In decentralised cryptocurrencies, the asset does not exist independently of the ledger maintaining it. A transfer recorded by the system constitutes the asset's transfer. No issuer redeems value, no administrator updates a register, and no court determines ownership as part of normal operation. Disputes concern only whether the system operated in accordance with its rules, not whether an external obligation must be honoured or recognised. The operative distinction of atomic tokenisation from other categories is the absence of any identifiable counterparty whose non-performance places the transaction at risk. Network validity, somewhat akin to referential tokenisation, is dependent on distributed agreement. In referential, contract, and title tokenisation a specific actor can fail, a counterparty may not follow through, and a registrar may not recognise the transfer. In atomic arrangements there is no actor that can trigger or contest an action. The network could malfunction, but no individual or entity can decide or default in the transfer of entitlement.

Attempts to apply the same logic to external assets do not presently achieve this result. Tokens referencing property, securities, or contractual rights ultimately depend on redemption, enforcement, or institutional recognition outside of the recording mechanism. The recorded state may signal or initiate a transfer, but it does not determine entitlement. Authority, therefore, remains external to the system even where automation is extensive. The difference is structural, not technological. When the asset and the record are identical, ownership is determined by operation. If the asset exists outside the system, the institutions governing that external reality determine ownership. Current tokenisation of real-world assets, therefore, cannot occupy this category, notwithstanding its frequent portrayal as such in both literature and practice.

These examples show that systems commonly described under a single term operate under fundamentally different conditions of completion. Stablecoins circulate claims, tokenised securities activate enforceable rights, recognised digital registers transfer acknowledged title, and only native digital assets transfer ownership by operation of the record itself. The shared terminology obscures differences in transfer implementation and outcome.

Rather than arising from theoretical disagreement, this difference reflects an unresolved practical problem. How can the recorded event be made to coincide with the operative transfer? The technical development of token standards can be read as a sequence of attempts to remove the remaining steps required after a ledger update for entitlement to become effective. These distinctions are not static. Across contemporary tokenisation practices, there is a discernible progression in how the transfer is completed where arrangement is dependent on externality to systems in which completion is integral within the ledger itself. The following section traces this progression, showing how successive designs internalise functions previously performed by external institutions.

The progressive internalisation of transfer

The difference mentioned above is both theoretical and practical. Systems described as "tokenised" frequently leave unresolved steps between a recorded transaction and an operative transfer. Market participants encounter this as operational friction. After ledger updates, redemption processes, transfer agents, registrars, compliance approvals, and legal enforcement are still required.

The development of Ethereum Request for Comments (ERC) token standards can be read retrospectively as a sequence of technical attempts to reduce this friction. The standards were not designed in coordinated progression because their authors addressed distinct, local problems rather than working from a shared theory of settlement or property completion. Practitioners iterated solutions to their immediate problems outside of a comprehensive framework that would allow for integrated understanding. Yet when viewed within this paper's analytical continuum, each successive design modifies the transfer mechanism in ways that increasingly align the recorded ledger event with the moment at which the transaction becomes operative. What appears historically as independent engineering can be analytically interpreted as movement along the conditions of completion.

ERC-20, proposed by Vogelsteller and Buterin in 2015, solved the problem of distribution rather than transfer completion (Vogelsteller & Buterin, 2015). It allowed units to be created, subdivided, and transferred between participants without maintaining individual ledgers for each issuer. However, the recorded movement of units did not itself complete a change in entitlement beyond the system maintaining the balances. The ledger resolved who held the token but not whether the holder possessed what the token referred to. ERC-20 thus internalised accounting but not settlement, remaining within referential tokenisation.

Developed by Entriken, Shirley, Evans, and Sachs in 2018, ERC-721, or the non-fungible token (NFT) assigned a unique identifier to each token to make the record non-fungible (Entriken et al., 2018). The recorded transfer now distinguished one unit from another and enabled tokens to point to specific things outside the system. However, the change only affected identification, not completion. The holder gained certainty about which token they possessed, but not whether possession of the token constituted possession of the underlying interest. ERC-721 made identification internal whilst leaving entitlement external.

As the externality problem became increasingly visible, new token standards emerged specifically designed for regulated instruments. Standards such as ERC-1400 and later ERC-3643 embedded legal, anti-money laundering, and regulatory compliance checks directly into the smart contract (Tokeny, 2023; Polymath, 2018). A token could only move where the legal constraints of the asset permitted it to move. This marked a change in function. The recorded event now activated obligations and legal consequences rather than describing a proposed change at its basic level of operation. However, completion still depended on recognition and enforceability outside the system. These implementations internalised validation but not authority, occupying contractual tokenisation.

Where compliance and eligibility conditions are embedded into transfer logic, the remaining external step is institutional recognition. A different arrangement emerges where the recognised ownership register is maintained through the token record itself. Under the Swiss framework noted above, a company may designate a blockchain register as its legal share register so that a transfer recorded on the authorised ledger constitutes the change in shareholder ownership (Lavayssière, 2025). Similarly, under the retrospectively mentioned German Electronic Securities Act, securities may exist solely as entries in an approved electronic register, with legal status determined by the register entry itself (BaFin, 2026). This stage internalises recognition but not the sovereignty of the transaction. These arrangements occupy title tokenisation.

If recognition itself is delegated to the record, the final remaining external step is the authority that grants the record its validity. The endpoint of the progression is therefore an arrangement in which no external institution determines ownership, and no further act is required after the recorded event. This condition exists today only where the asset is native to the system maintaining the record. In decentralised cryptocurrencies, the asset does not exist independently of the ledger that records it. Attempts to apply this logic to external assets do not currently achieve this effect. Tokens referencing property, securities, or contractual rights ultimately depend on redemption, enforcement, or recognised legal authority beyond the transfer system in which they are traded (North, 1990).

The progression of standards thus converges on a condition that only digital-native assets currently satisfy. ERC-20 internalised accounting while leaving entitlement external, ERC-721 internalised identification without altering authority, ERC-1400 and ERC-3643 embedded validation and enabled enforceable transfer, recognised digital registers delegated title to the recorded state, and native digital assets presently achieve transfer by operation alone. This evolution does not represent successive versions of the same mechanism but successive attempts to remove the remaining institutional step after a recorded event. What appears as technical iteration is actually institutional migration. Honouring gives way to enforcement, enforcement to recognition, and recognition approaches operation. The taxonomy described above is not a classification imposed upon the technology but a structure that the technology itself has been approaching without intentional organisation.

The atomic tokenisation of real-world assets remains elusive because the surrounding authority structure remains external to the record, not because the technology is immature. Technical standards can validate, calculate, and restrict transfer, but ownership changes only where a legal or social system accepts the recorded state as decisive (Wandhöfer & Berndsen, 2019; Liao, 2020). Until that point, tokenisation improves administration rather than settlement.

Developments such as ERC-4626 vault tokens and dynamic compliance proposals, including ERC-7518, do not introduce further categories for analysis but compress the remaining distance between contractual enforcement and recognised title by internalising additional post-transfer functions. Emerging proposals such as ERC-7208 with on chain data containers that hold much promise for real world asset tokenisation, further internalise persistent asset data, decoupling storage from logic to code seamless adaptation across token standards without additional migration. Recent standards follow this trajectory by incorporating accounting, redemption logic, and adaptive eligibility directly into the transfer mechanism. The recorded event increasingly performs tasks previously carried out by administrators, agents, or counterparties. But these developments do not eliminate the finality of institutional requirements. The operative change still depends on whether an external system treats the recorded state as authoritative. Technological progress on ERC token standards continues toward completion, without specifying when it will occur. The boundary remains institutional, not technical.

Tokenisation evolves by relocating institutional functions into the moment of record, where completion relies on the location of its governing authority. The difference between implementations is not the presence of a token but the source that determines entitlement once the token moves.

The regulatory implications of this taxonomy are significant. For regulators and policymakers, classifying tokenised systems by operative authority provides a practical tool for identifying and addressing legal or systemic risks. This framework allows regulators to distinguish between arrangements that only represent exposure and those that truly effectuate ownership, enabling tailored regulatory responses based on the degree of external dependence or embedded authority. For instance, rules around investor protection, disclosure, and liability can be calibrated to the specific point at which entitlement becomes operative. Where completion remains external, regulators may require robust disclosure of counterparty, administrative, and jurisdictional risks, as well as the enforceability of claims and the availability of dispute-resolution mechanisms. For systems approaching title or atomic arrangements, policy might focus on clarifying the legal status of digital records, establishing procedures for recognition, and ensuring the resilience of the underlying infrastructure. By mapping tokenised implementations onto this taxonomy, regulators can better identify gaps in legal recognition, investor safeguards, or systemic oversight, and direct their interventions to the precise institutional layer that determines completion. This taxonomy therefore enhances regulatory clarity and provides a concrete basis for effective risk management as tokenisation technology continues to advance.

Market consequences of completion authority

The implications of completion authority do not appear at once but reveal themselves progressively. What first appears as a question of trading becomes a question of settlement, then of valuation, and finally of authority. The familiar critiques directed at tokenisation arise when these stages are discussed independently of the structural condition that produces them. Considered together, they are not separate problems but successive expressions of the same distinction. The framework predicts four observable consequences. First, liquidity should be inversely proportional to the degree of real-world finality embedded in the transfer. Second, systems where transfers do not themselves complete entitlement cannot function as settled base assets within composable environments. Third, the visible reliability of the ledger should mask the invisible variability of authority, leading to systematic misclassification of risk. Fourth, governance without jurisdictional reach is coordination, not control. Each pattern is observable in current tokenised markets.

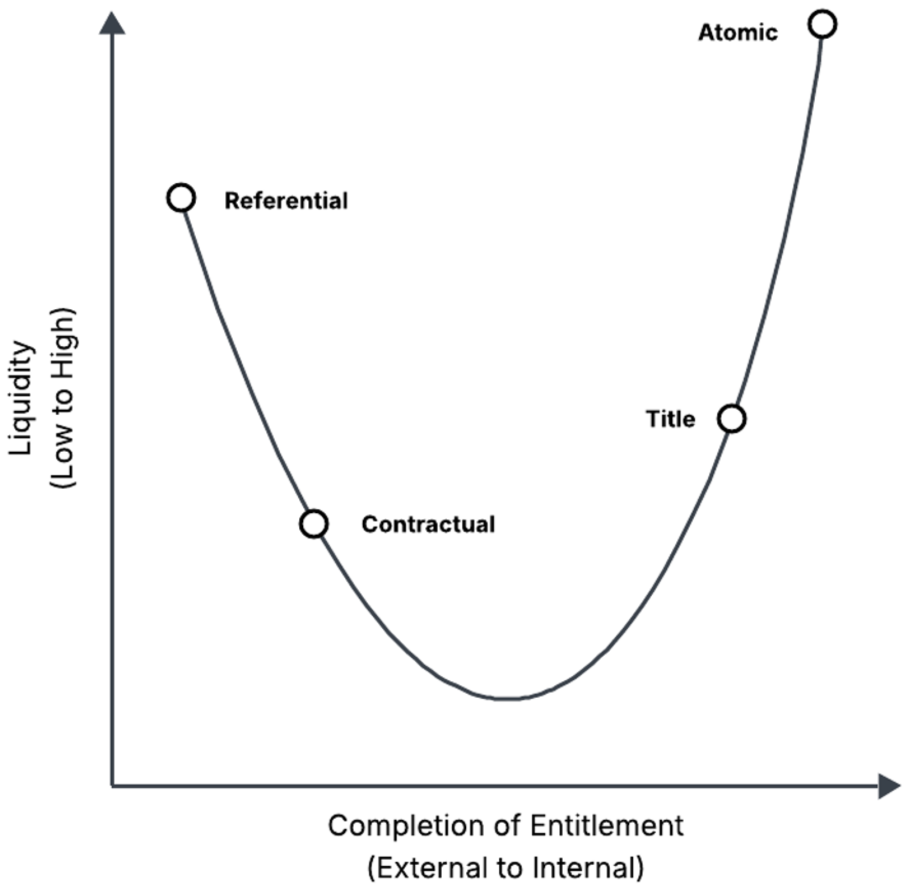
Liquidity and completion

Tokenisation is routinely associated with increased liquidity. Fractionalisation, continuous trading environments, and global access are suggested as mechanisms through which previously illiquid assets become freely exchangeable in digital form. However, secondary markets for many tokenised assets remain persistently thin despite technical tradability and the broad and global access that crypto markets are lauded for. Mafrur (2025) states this plainly,

"While nearly any asset can be technically represented as a token on-chain, ensuring those tokens are actively tradable remains a far more difficult challenge… Most RWA tokens show low trading volumes, long holding periods, and limited investor participation, even with 24/7 global market potential… secondary trading remains limited [across over $25 billion in tokenized RWAs]."

This is not simply a delay. Latency postpones a result that is already determined. Dependency leaves the result contingent. A market can tolerate latency and remain liquid if the outcome is assured. It cannot, however, ignore dependence where the trade's effect is conditional on additional cooperation, recognition, or enforcement (FSB, 2024). Participants will therefore not only price the asset, but also the process dependencies required to make ownership effective. Spread and discount emerge not because the token cannot be traded, but because what is acquired is partly a claim upon completion rather than completion itself. The framework predicts a systemic relationship between liquidity and the location of completion authority illustrated in Figure 2.

Figure 2: Liquidity as a function of completion



Note: Liquidity declines as entitlement becomes institutionally operative, reflecting dependence on external recognition or enforcement, before rising again where completion becomes endogenous to system operation.

To be accurate, liquidity follows completion rather than representation. Where the transfer settles the position entirely, participation expands because exit requires no further involvement. Where completion remains external, participation contracts because exit depends on the remaining process being carried through by third-party actors. Where completion lies beyond the recorded transfer, markets exchange exposure. Where completion lies within it, they exchange ownership. Liquidity is therefore inversely proportional to the degree of real-world finality embedded in the transfer of tokenised assets (Wandhöfer & Berndsen, 2019).

Settlement and composability

Tokenised systems are frequently described as providing instant settlement. On-chain transfers execute within seconds, balances update deterministically, and counterparties no longer wait for clearing cycles. From the perspective of the ledger, the transaction is complete. Yet settlement in economic terms concerns the resolution of entitlement and not the basic function that records a movement of a token. A transfer settles only when no further act is required for the parties' positions to become operative (BIS CPMI, 2017; Liao, 2020).

The common description of tokenisation as instant settlement conflates finality of record with finality of entitlement (FSB, 2024; BIS, 2023). In referential and contractual categories, the technology removes reconciliation within the ledger but does not necessarily remove the institutional processes required to make the transfer effective. What has accelerated is confirmation, not completion.

The consequence is practical rather than semantic. Instant settlement is an illusion. Systems where transfers do not themselves complete entitlement cannot function as settled base assets within composable environments. A protocol may treat the recorded balance as final while the economic position remains contingent on processes outside the system. What appears atomic becomes conditional, and what appears delivered can only ever be promissory. Native digital assets compose because the transfer resolves entitlement. Tokenised assets resist composition because transfer only initiates a process that is concluded elsewhere. The boundary encountered in tokenised asset management is the threshold between completed and uncompleted transfers, not a technical issue between blockchains and institutions.

Risk misclassification

When systems that operate under different conditions of completion are described using a single term, risk is assessed as though the underlying mechanisms were equivalent. Participants evaluate tokenised assets according to surface properties such as transfer speed, programmability, or fractional access, while the actual exposure depends on where entitlement becomes operative. The result is a systemic misclassification of risk rather than a simple underestimation of it (FSB, 2024; IOSCO, 2025).

The consequence of risk misclassification is an inversion of perception. The visible reliability of the ledger masks the invisible variability of authority. Technical finality is mistaken for legal finality, and operational determinism is mistaken for economic certainty. Markets, regulators, and designers therefore analyse identical tokens through incompatible frameworks, each internally consistent yet collectively contradictory.

This misperception is particularly acute in the tokenisation of assets, where investors may potentially and unknowingly treat the token as direct property despite uncertainty in the underlying entitlement. As both the CFA Institute (2025) and IOSCO (2025) note, despite owning the digital token, investors may not understand the legal aspects of ownership rights and their rights to transfer tokenised financial assets, especially in hybrid arrangements common to real-world assets. This illusion alters withdrawal behaviour because when confidence in the intermediary weakens, participants demand delivery of the instrument that cannot be provided, converting contingent claims into sudden stress rather than progressive correction. Failures in such systems therefore appear abrupt because perceived entitlement and actual entitlement diverge at the critical moment (FSB, 2024; Garrido, 2023).

Governance and authority

The distinction between forms of tokenisation becomes visible at intervention after purchase. A holder may trade, price, and collateralise a token while assuming entitlement to the underlying asset. The assumption is tested only when the holder attempts to act upon potentially assumed power or rights rather than exchange. At that moment, the nature of the position becomes clear. The holder discovers whether they possess authority over the asset or just a basic claim whose outcome depends on another party.

Tokenised systems are often described as decentralising control by attaching decision rights to transferable tokens. Voting mechanisms and automated execution appear to allow holders to govern the underlying asset (Cong et al., 2025). The assumption is that possession of the token carries authority. In practice, governance follows completion of the transfer. What a holder can determine depends on what the transfer actually moved (Eisermann et al., 2025).

Governance without jurisdictional reach is coordination. It is not control. Where the asset's state can change independently of the system, governance expresses preference. Where the asset's state exists only within the system, governance determines outcome. Tokenisation standardises the appearance of governance while varying the location of authority. Identical mechanisms can therefore represent discussion, instruction, or control depending on what the transfer conveyed. The appearance of decentralised governance arises from observing a common procedure across different authorities (Appel & Grennan, 2023; Cong et al., 2025; Eisermann et al., 2025).

Table 2: Summary of tokenisation taxonomy

Attribute	Referential	Contractual	Title	Atomic
What the token moves	Exposure	Enforceable claim	Legal ownership	Ownership itself
Completion authority	Honour / performance	Legal enforcement	Legal recognition	System operation
Ledger role	Mirror	Execution trigger	Official register	Mechanism of transfer
Entitlement change occurs	Outside system	After adjudication	At recognised record	At execution
Settlement finality	Redemption	Court / administrator	Registry acceptance	Protocol consensus
Governance effect	None	Instruction	Binding	Intrinsic
Risk locus	Issuer solvency	Legal enforceability	Institutional validity	Protocol correctness
Liquidity expectation	High but fragile	Moderate	Restricted	Native
Composability	Artificial	Conditional	Limited	Autonomous
Failure mode	Default	Dispute	Jurisdiction conflict	Software fault
Loci of control	Issuer	Legal operator	Registered holder	Key holder
Court determines ownership from	Contract	Contract	Register	Chain
Example structures	Stablecoins, synthetics	SPV tokens, funds	DLT securities	Native crypto
Typical token standards	ERC-20 wrappers	ERC-1400 family	ERC-3643 registers	Native protocol

Tokenisation can distribute coordination without relocating control. The presence of voting does not demonstrate the asset's decentralisation, but it does democratise the decentralisation of expectations about it. Governance follows the nature of the entitlement transferred. If authority does not move, the transfer concerns a relationship to the asset rather than ownership.

Governance, therefore, functions as a diagnostic. Liquidity reveals how positions trade, settlement reveals when they resolve, risk reveals what they expose, and governance reveals what they confer. Where decisions cannot alter the asset's state, the recorded transfer did not transfer authority, regardless of how completely it relocated the token.

Conclusion

This discussion began with the observation that sovereign actors, financial institutions, and technology platforms discuss tokenisation as though it denotes a single economic transformation, when in reality it is far from this state of affairs. The contradictions that the paper presents, between promised and observed liquidity, recorded and operative settlement, technical finality and legal uncertainty, the appearance of governance and the reality of control (Appel & Grennan, 2023), are not the result of early markets or inadequate regulation. They are the result of analysing structurally different arrangements under a single, loose-categorising container that is potentially dangerous in its outcomes.

This paper has argued that the term tokenisation currently collapses together arrangements that perform fundamentally different institutional functions. Some circulate exposure, some activate enforceable rights, some transfer recognised title, and only a limited subcategory of tokenisation transfers ownership by operation of the record itself. The contradictions seen across liquidity, settlement, risk, and governance arise from analysing different completion conditions under a single name rather than disagreements about the same phenomenon of "tokenisation". Each disciplinary lens, whether legal, financial, or technical, reaches internally consistent conclusions precisely because it is examining a different object from its own disciplinary silo. The apparent tautology dissolves once the object is specified.

Rather than defining tokenised systems by technological architecture, asset type, or regulatory category, this paper proposes a taxonomy that classifies them according to the point at which entitlement becomes operative and the institution that determines that operability. By distinguishing referential, contractual, title, and atomic arrangements, the analysis offers a single variable that explains divergent observations across tokenised markets. The relevant distinction is based on whether any system of authoritative engagement is still necessary after the recorded transfer. The distinction is not reduced to either technological architecture or asset type. Where redemption, enforcement, or recognition is still required, the token relocates a relationship. Where no further action is required, the token relocates the asset. Network state-style considerations illustrate a possible state in which assets may cease to be external to the systems that record them. In such environments, governance, membership, and economic participation are baked within a single institutional layer. Where the asset and the authority determining its ownership exist in tandem, the conditions for constitutive transfer may become conceivable (Srinvasan, 2022). These models do not yet achieve atomic tokenisation of external assets, but they indicate how real-world assets might be structured so that economic, legal, and recorded states converge rather than reconcile. This clarifies why market behaviour appears inconsistent. Liquid trading can coexist with uncertain ownership. Rapid confirmation can sit alongside incomplete settlement. Operational determinism can mask institutional dependency. Distributed participation can leave authority entirely unchanged. These are errors of treating arrangements with different operative moments as equivalent. They are not errors of tokenisation.

The risks arising from this categorical imprecision are semantic rather than technological in origin. A lack of awareness of the categorical distinctions underlying tokenisation breeds confusion when a single term is used to encompass fundamentally different arrangements. Participants frequently assume the properties of one form of transfer while interacting with another. The framework offered here seeks to address these classification misunderstandings without prescribing specific implementations. It provides a common reference point through which implementations can be evaluated without collapsing their differences into either unchecked technological optimism or blanket scepticism.

Compared to existing academic frameworks, this taxonomy moves beyond classifications based solely on asset class, technological protocol, or regulatory status. Previous approaches, such as those in Zetzsche et al. (2020), typically distinguish tokenised assets by reference to legal forms or regulatory categories, while the BIS (2023) and OECD (2024) frameworks often focus on use cases, infrastructural roles, or technological layers. By contrast, the framework developed here organises tokenised systems according to the moment when entitlement becomes operative and the source of completion authority. This focus on operative effect and institutional dependency distinguishes the present taxonomy from prior models, providing a unified analytical tool for comparing systems that would otherwise fall under different categories. In doing so, it enables a more precise assessment of legal risk, market function, and regulatory challenges across a broad spectrum of implementations. Future discussion of tokenised systems can therefore proceed by specifying what the transfer completes rather than assuming what the presence of a token implies.

Tokenisation's significance will ultimately depend which institutions accept the recorded state as decisive so the market can price the friction thresholds and speed of resolution appropriately. This framework makes that progress intelligible by distinguishing when a system changes representation and when it changes ownership.

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